

A Packing of 19 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 19 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.686483. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.678978, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 19 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.686483 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 19$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.678978, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 19/18 = 2.686869$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N19.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
391615742392440004889334436609434463730735360884096033429709055862210042
205300368438687710785496261215157598322328892111309536364723914046162270
91148925021030287730315841681362195803715648
```

denominator:

```
145772594752234636525515659044568981117065731453040768085553975194308834
581049459289576712014515372687075320377986609590940375626752703983407342
85845234489616635750553266025256972903788103
```

B Coordinates

19 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N19.txt`.

```
353148017133 143921556017 -47112758863 99160622353 -482440191542 398567841070 135824198758 -485161064352 603706143 155791297883 497446136531 153497172708
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